Question				Marking details	Marks Available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
1	*(a)*	(i)		Ticks in boxes 1, 5 and 6 Television 1 uses less energy per second than television 2 ✓ Television 3 uses 40 units more per year than television 4 ✓ Televisions with the same energy rating, e.g A++, don't always have the same power ✓ –1 mark for each additional box ticked		3		3		
	(b)			**N.B. Only televisions 1 and 2 to be used.** **1st mark – correct substitution of one ratio** **2nd mark – correct calculation of one ratio** **3rd mark – correct calculation of 2nd ratio** **3 marks to be awarded only if correct conclusion present** Screen size to screen size compared with power to power $\frac{139}{69} = 2.01 \quad \frac{78}{32} = 2.44$ **OR** $\frac{69}{139} = 0.50 \quad \frac{32}{78} = 0.41$ Conclusion – [Ratios not the same] so not true **Alternative** Ratio of screen size to power compared $\frac{69}{32} = 2.16 \quad \frac{139}{78} = 1.78$ **OR** $\frac{32}{69} = 0.46 \quad \frac{78}{139} = 0.56$ Conclusion – [Ratios not the same] so not true **Alternative** Ratio of screen size to kWh per year compared $\frac{69}{47} = 1.47 \quad \frac{139}{108} = 1.29$			3	3	3	

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				OR $\frac{47}{69} = 0.68 \qquad \frac{108}{139} = 0.78$ Conclusion – [Ratios not the same] so not true **Alternative:** Screen size to screen size compared with kWh per year to kWh per year $\frac{69}{139} = 0.50 \qquad \frac{47}{108} = 0.44$ **OR** $\frac{139}{69} = 2.01 \qquad \frac{108}{47} = 2.30$ Conclusion – [Ratios not the same] so not true						
	(c)	(i)		Time = $\frac{108}{\left(\frac{78}{1000}\right)}$ (1) substitution [even for $\frac{108}{78}$] Time = 1 384.6 [hours] (1) correct answer correctly rounded Answer = 1.38×10^{n} where n is not 3 award 1 mark only	1	1		2	2	

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Cost = 108 × 16 **or** 108 × 0.16 (1) substitution Cost = £17.28 (1) answer Accept £17 or £17.00	1	1		2	2	
		(iii)		Running cost of TV 2 for 10 years = £17.28 **(ecf)** × 10 = £172.80 (1) Accept £170 or £172 or £173 Running cost of TV 4 for 10 years = 172 × 10 × 0.16 = £275.20 (1) Accept £275 TV 4 costs £102.40 more to run but it is £200 cheaper to buy so Sarah is right (1) **Alternative:** Annual savings from using TV 2 = (172 - 108) × 0.16 = £10.24 (1) Running cost = £10.24 × 10 = £102.40 (1) **OR** Difference in units over 10 years (172 - 108) × 10 = 640 (1) Difference in running cost = 640 × 0.16 = £102.40 (1) **3rd mark -** TV 4 costs £102.40 more to run but it is £200 cheaper to buy so Sarah is right (1) **Alternative:** Total cost of TV 2 = £172.80 **ecf** (1) + £1 000 = £1 172.80 (1) Total cost of TV 4 = £1 075.20 so cheaper so Sarah is right (1) **OR** Total cost of TV 4 = £275.20 (1) + £800 = £1 075.20 (1) Total cost of TV 2 = £1 172.80 so more expensive so Sarah is right (1) **Alternative:** Annual savings from using TV 2 = (172 - 108) × 0.16 = £10.24(1) Payback time = $\frac{200}{10.24}$(1) = 19.5 years which is longer than 10 years so Sarah is right (1)			3	3	2	

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		Fewer power stations / less fuel burnt (1) Reducing CO_2 emissions / reducing impact on global warming / reduces carbon footprint (1)	2			2		
				Question 1 total	**4**	**5**	**6**	**15**	**9**	**0**